Mini Project 1: Setup and Build Your Own Cybersecurity Lab

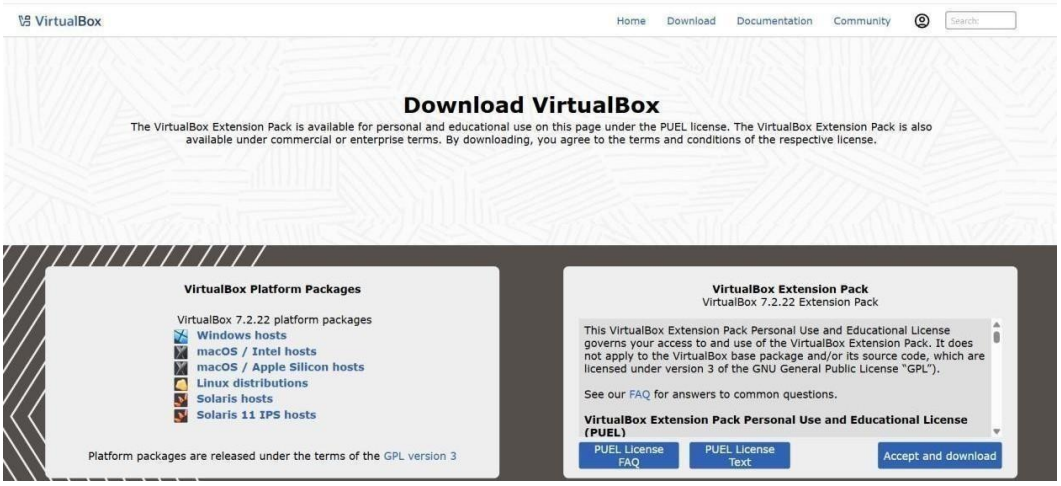
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Course / Group	Cyber Security / G17
Platform	SURE ProEd
Host OS	Windows 11
Date	24/10/2026

1. Aim

To prepare a safe, isolated practice environment by installing VirtualBox on Windows 11 and running Kali Linux inside a virtual machine.

Step 1: Download VirtualBox

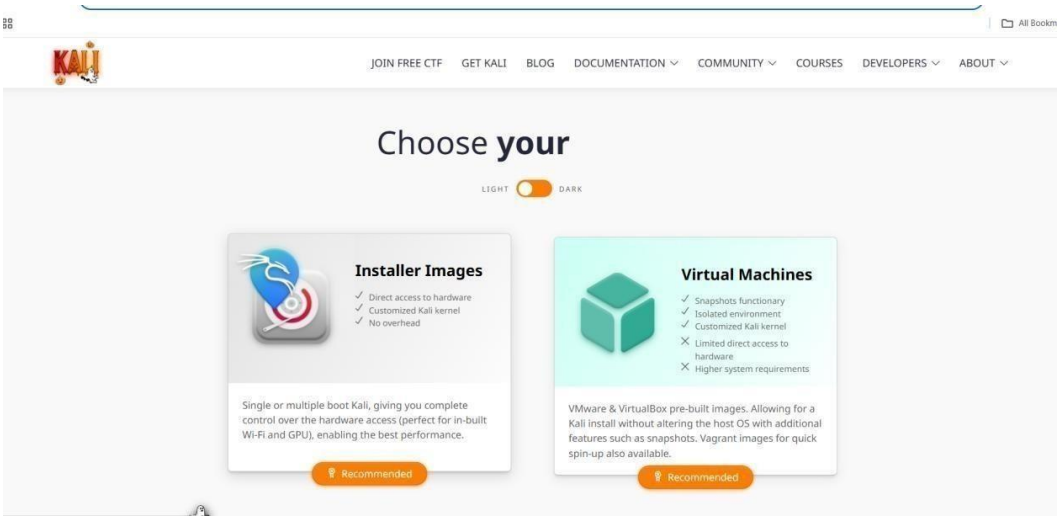
Open Google Chrome, search for VirtualBox and go to the official downloads page.
https://www.virtualbox.org/wiki/Downloads



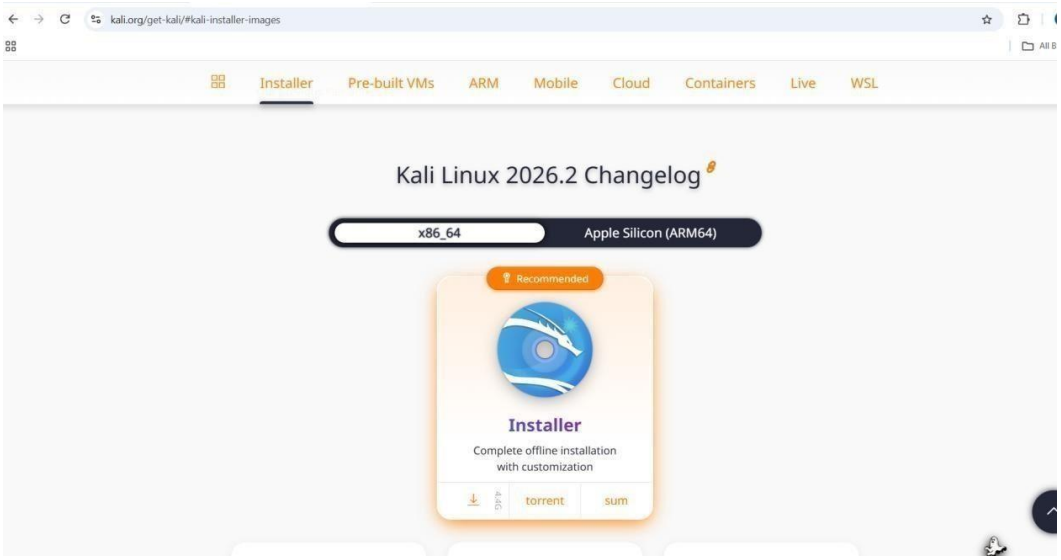
Choose the installer for Windows hosts. Also download the Extension Pack from the same page; the download starts as soon as the option is selected. The version used in this lab is **VirtualBox 7.2.20** with the matching **Extension Pack 7.2.20**. The Downloads page may show a newer build number than the installer used here; the installer and Extension Pack must always have the same version.

Step 2: Download Kali Linux

Search for Kali Linux and open the official download page. https://www.kali.org/get-kali/#kali-platforms



The page lists several platforms. Kali offers ready-made virtual machine images, which are the quickest option. In this lab the **Installer image (ISO)** was used instead, so the virtual machine is created and installed manually in VirtualBox. https://www.kali.org/get-kali/#kali-installer-images

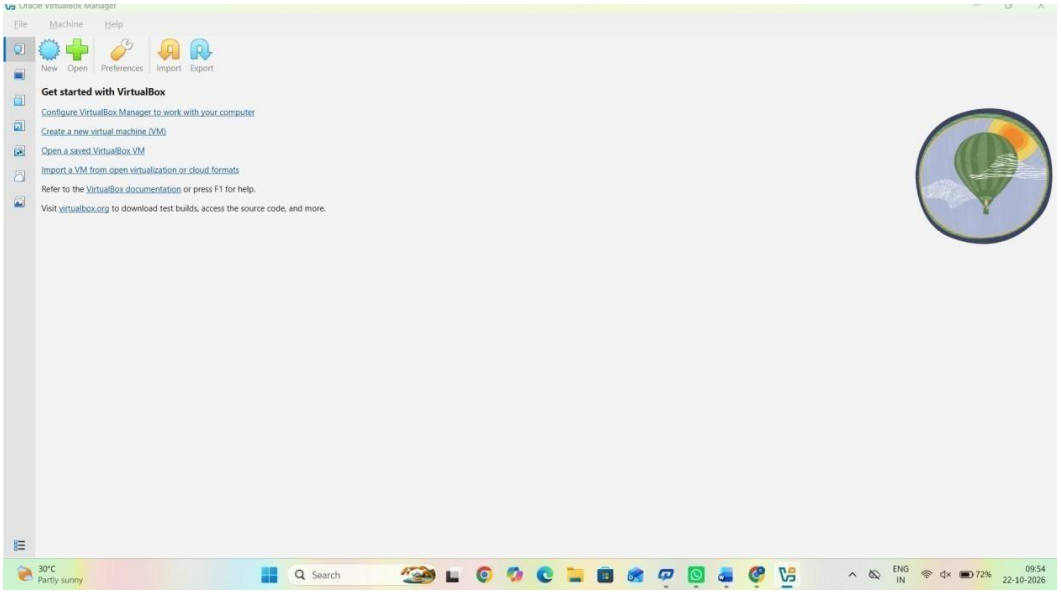


Select the x86_64 Installer image. Once the download finishes, the file (*kali-linux-2026.2-installer-amd64.iso*) appears in the browser downloads list.

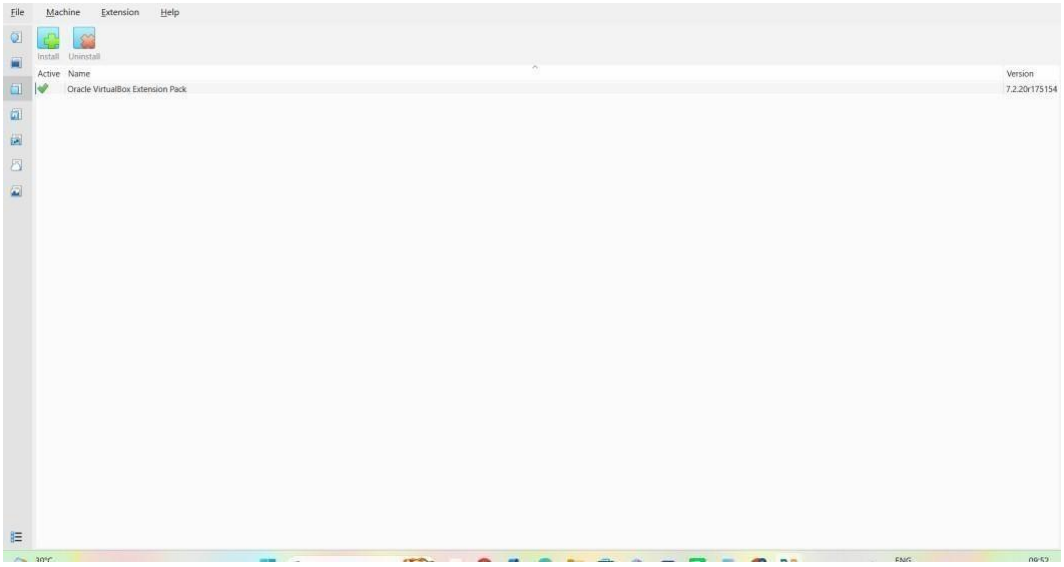
Step 3: Install VirtualBox and the Extension Pack



Open File Explorer, run the VirtualBox 7.2.20 installer and keep clicking Next, accepting the defaults, until the installation completes.

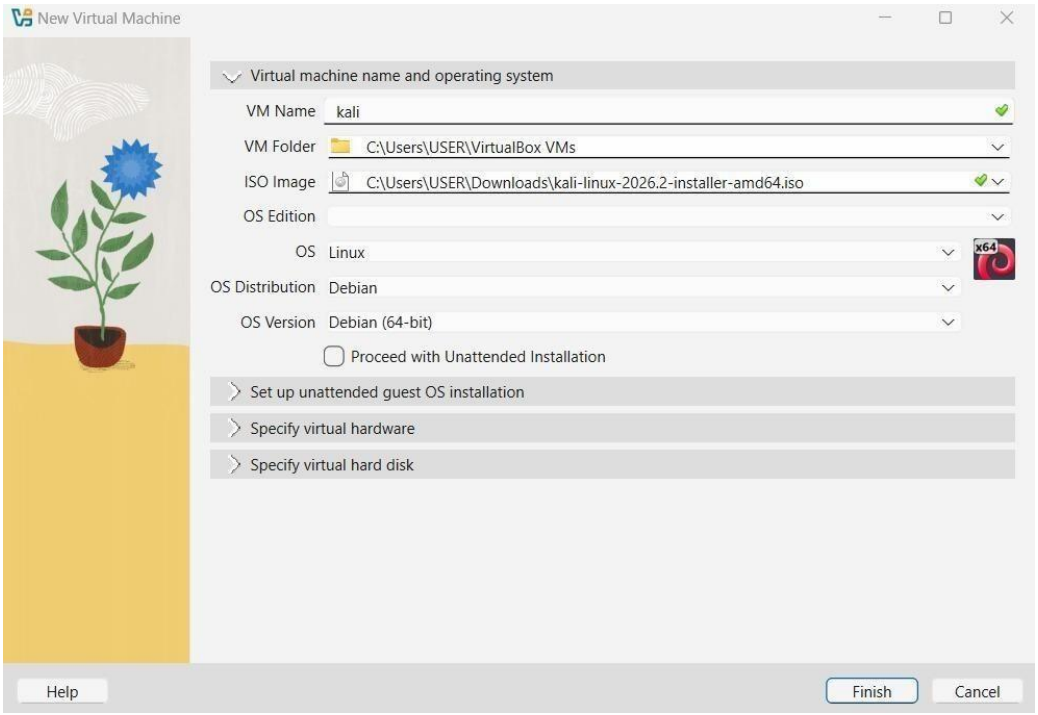


Then open the Extensions option in VirtualBox and select the downloaded Extension Pack file (7.2.20) to install it. The window below confirms the extension was added.

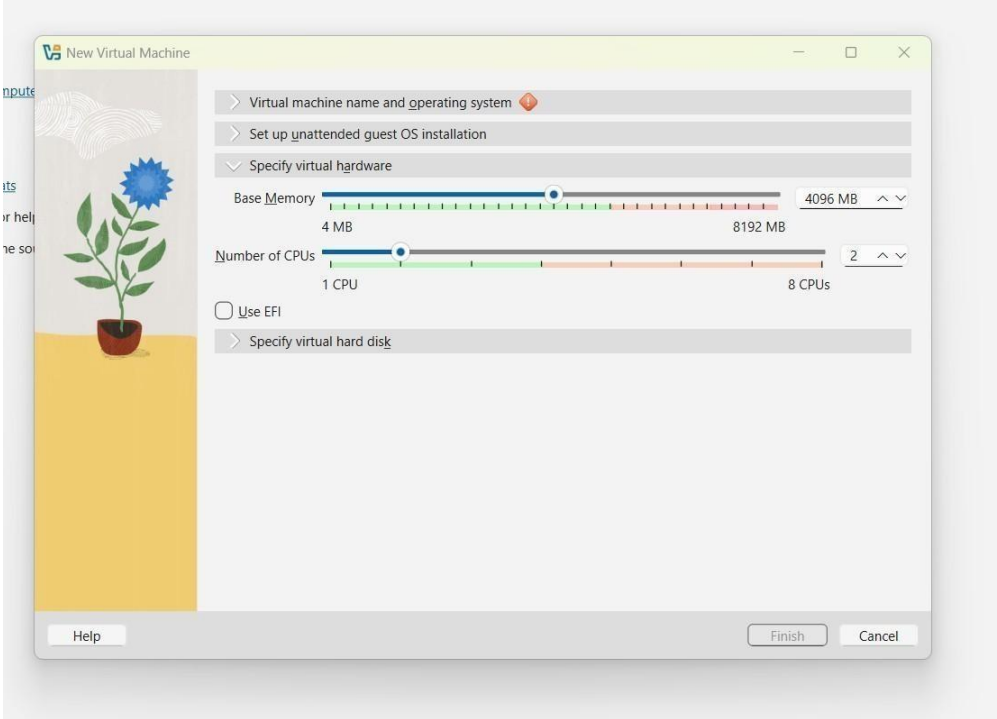


Step 4: Create the Virtual Machine

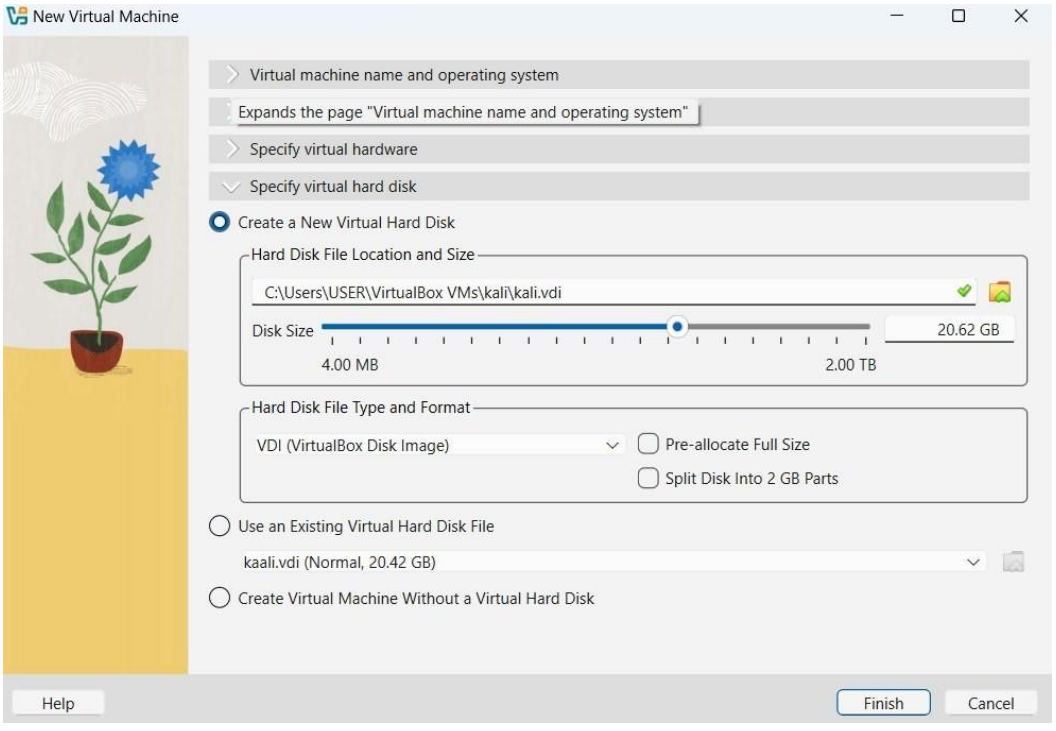
From the VirtualBox home screen click New to start the wizard. Enter the virtual machine name, select the Kali ISO image, and set the OS, distribution and version as shown in the screenshot. Leave **Proceed with Unattended Installation** unchecked, because Kali is installed manually with its own installer.



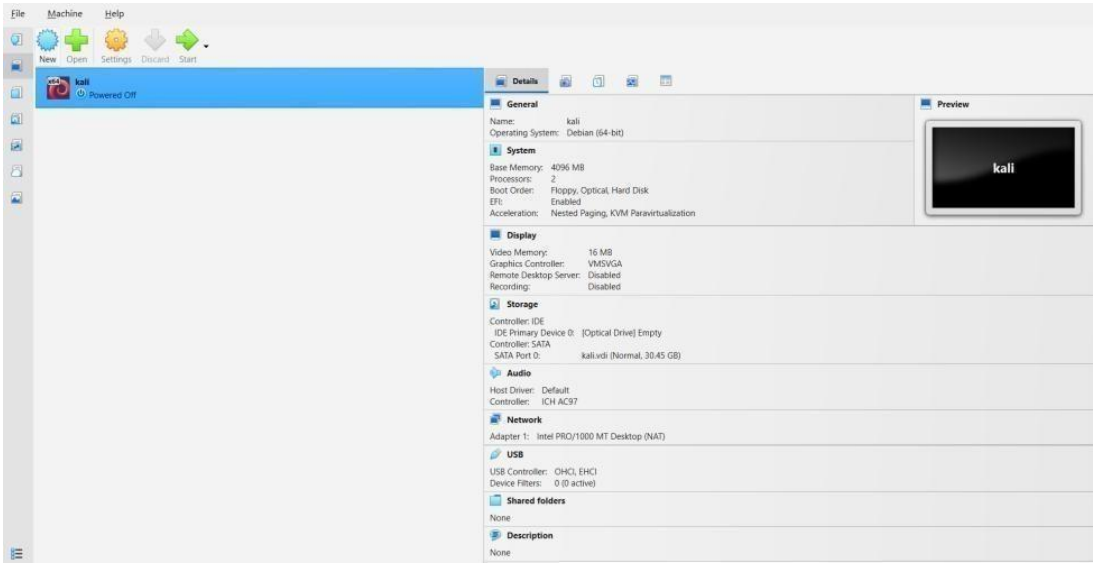
Under *Specify virtual hardware*, set the base memory to 4096 MB and 2 CPUs.



Under *Specify virtual hard disk*, create a new VDI virtual hard disk of about 20.6 GB. The virtual disk file was created as *kaali.vdi*, as shown in the screenshot, and it is attached to the virtual machine named *kali*.



When the Finish button becomes active, click it. The new virtual machine appears in the VirtualBox Manager in the powered-off state.

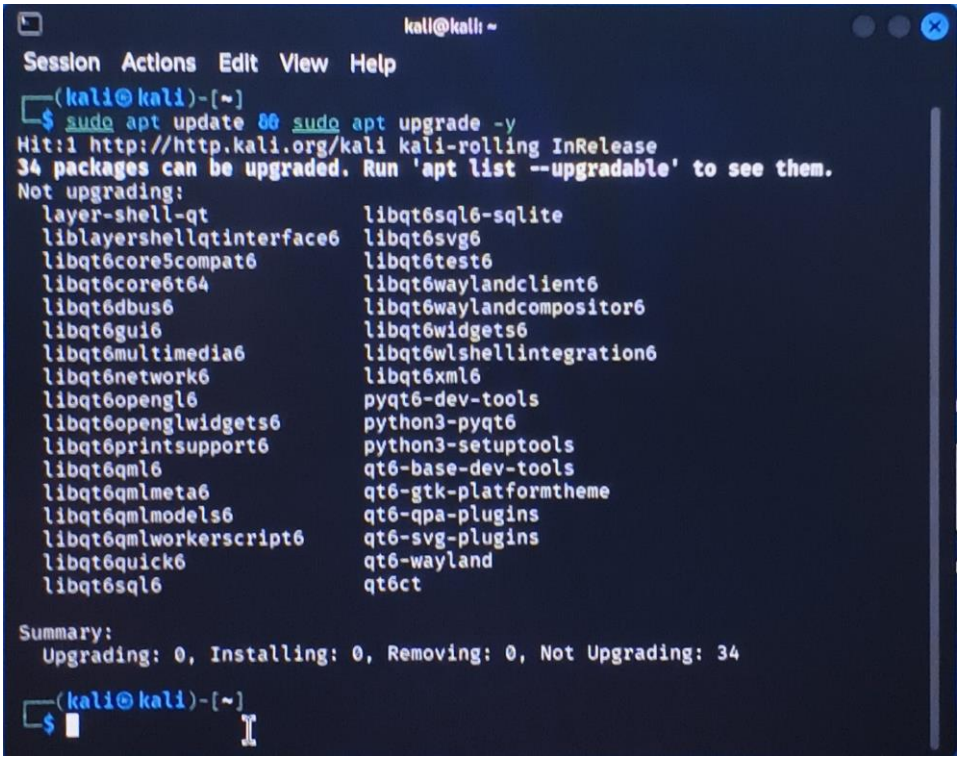


Step 5: Start Kali Linux, Install It and Update It

Click Start to power on the virtual machine. The Kali installer boots from the ISO; follow the installer steps (language, location, keyboard, network, partitioning) and create your own username and password when asked (kept confidential here). After the installation finishes and the machine restarts, log in with the username and password created during installation.

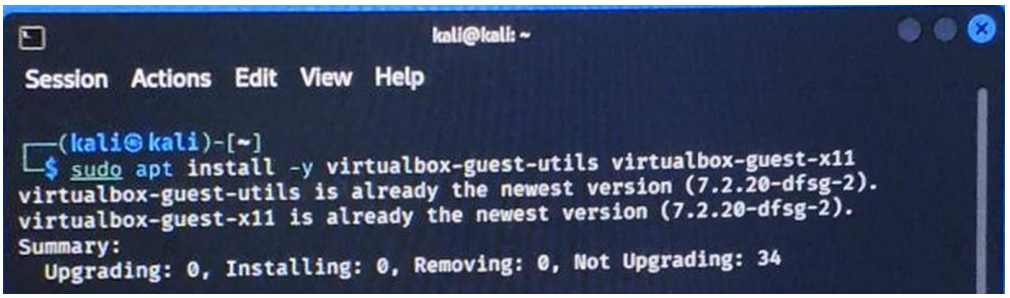
Open the Kali terminal and update the system with the following command:

```
sudo apt update && sudo apt upgrade -y
```



Next, install the VirtualBox guest additions so that screen resizing and clipboard sharing work:

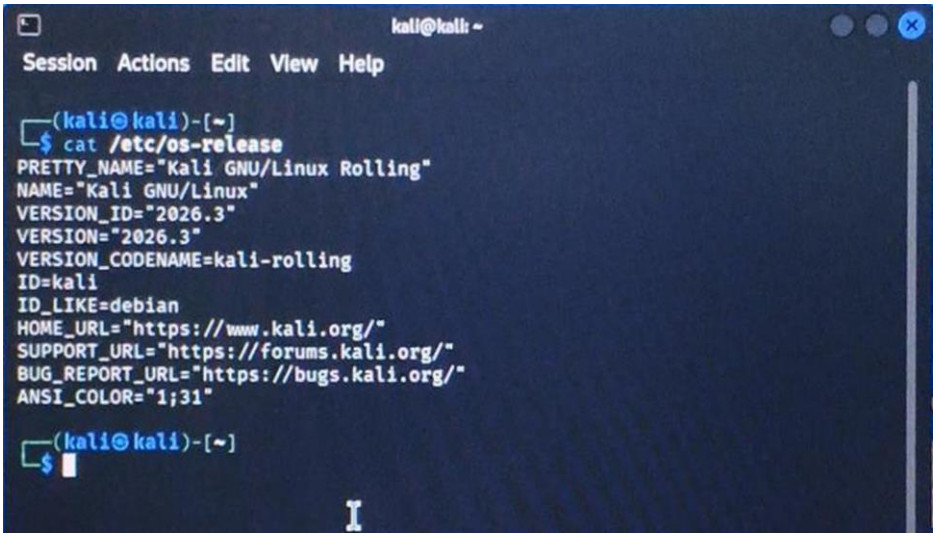
```
sudo apt install -y virtualbox-guest-utils virtualbox-guest-x11
```



Step 6: Verify the Installation

Check the operating system details with the command below. The output shows Kali GNU/Linux Rolling, version 2026.3. The ISO downloaded in Step 2 was 2026.2; because Kali is a rolling release, the system update in Step 5 brought it to 2026.3. This confirms that Kali is installed and up to date.

```
cat /etc/os-release
```



Restart the virtual machine to apply the changes and log in again to reach the Kali desktop.

Final Login

After the restart, the Kali login screen appears again. Sign in with the same details:

Username	Enter the credentials created during Kali Linux installation.
Password	Enter the credentials created during Kali Linux installation.

Once you are logged in, the cybersecurity lab is ready to use.

Conclusion

Kali Linux now runs as a virtual machine on VirtualBox 7.2.20 in Windows 11, with 4 GB of memory and 2 CPUs. It was installed from the Kali installer ISO (2026.2), updated to Kali 2026.3, fitted with guest additions and checked through the os-release output. The lab stays separate from the host system, so it is ready for the practical modules that follow.